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Efficacy and safety of anticoagulant prophylaxis to prevent venous thromboembolism in acutely ill medical inpatients: a meta-analysis

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Abstract. Själander A, Jansson J-H, Bergqvist D, Eriksson H, Carlberg B, Svensson P (Sundsvall Hospital, Sundsvall; Skellefteå Hospital, Skellefteå; Uppsala University Hospital, Uppsala; Sahlgrenska University Hospital, Östra, Göteborg; Umeå University Hospital, Umeå; and Malmö University Hospital, Malmö; Sweden). Efficacy and safety of anticoagulant prophylaxis to prevent venous thromboembolism in acutely ill medical inpatients: a meta-analysis. *J Intern Med* 2008; **263**: 52–60.

Objectives. Venous thromboembolism (VTE) is a potentially serious complication of hospitalization and immobilization. The use of anticoagulant prophylaxis in acutely ill medical inpatients is still under debate. New data including a recent meta-analysis have recently been published. We aim at studying the efficacy and safety of anticoagulant prophylaxis in acutely ill medical inpatients, and demonstrate differences between meta-analyses due to different data extraction from the heterogeneous studies included.

Subjects. The Cochrane Library, MEDLINE and EMBASE were searched from 1980 to present. Manual searches were performed regarding abstracts from major meetings. Seven blinded randomized controlled clinical trials assessing the prophylactic effect of heparin in acutely ill medical patients were identified and included in the meta-analysis.

Results. Low-molecular weight heparin (LMWH) prophylaxis prevented 48% of symptomatic pulmonary embolism (PE), 48% of symptomatic deep vein thrombosis (DVT) (not significant) and 51% of asymptomatic DVT. A nonsignificant trend towards higher bleeding risk during LMWH prophylaxis was found. Death was not significantly affected. We compared our data with a recent meta-analysis with different study selection and data extraction and found similar results.

Conclusions. As DVT and PE are manifestations of the same illness, VTE, one can argue that anticoagulant prophylaxis prevents approximately half of the expected events. Most medical inpatients have short hospital stays, and a low risk of VTE. The important task for the clinician is to identify patients with a sufficiently high risk of symptomatic VTE to warrant LMWH prophylaxis. Despite differences in study selection and data extraction, our study shows results similar to a recent meta-analysis.

Keywords: heparin, low-molecular weight heparin, prevention, pulmonary embolism, venous thromboembolism, venous thrombosis.

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Introduction

Venous thromboembolism (VTE) is not an uncommon complication of hospitalization and immobilization. The rationale for thromboprophylaxis in orthopaedic and general surgery patients is well documented [1, 2]. The risk period is relatively well defined, and the incidence of symptomatic VTE is high prompting the clinician to use VTE prophylaxis. In acutely ill medical patients, VTE prophylaxis is not widely used despite consensus statements recommending such patients to be assessed for risk of VTE on admission to the hospital and receive thromboprophylaxis where appropriate [2, 3]. The reluctance to implement thromboprophylaxis in acutely ill medical patients may reflect the difficulty to select high risk patients [4], and lack of data showing that symptomatic VTE can be prevented. A previous meta-analysis [5] consists of a limited number of studies, heterogeneous regarding patient populations and diagnostic procedures for VTE. Since then three large randomized studies have been published [6–8], each showing significant effect of prophylaxis with low-molecular weight heparin (LMWH) or fondaparinux on asymptomatic deep vein thrombosis (DVT) but not symptomatic VTE (Figs 1 and 2). Recently, a new meta-analysis has been published on this subject [9]. Although using the same search criteria as in our study, our results differed in some aspects. This is in itself interesting and will be discussed more in detail below. The aim of this study was to compare the efficacy and safety of unfractionated heparin (UFH) and LMWH including pentasaccharides versus placebo in the prevention of VTE in acutely ill medical inpatients.

Methods

Data sources

A literature search was performed to identify all published blinded randomized controlled trials on prophylaxis against VTE with UFH, LMWH or fondaparinux in acutely ill medical inpatients. The electronic databases MEDLINE, EMBASE and Cochrane database of clinical trials from 1980 to present were used. Articles were searched until October 2006. Manual search was performed

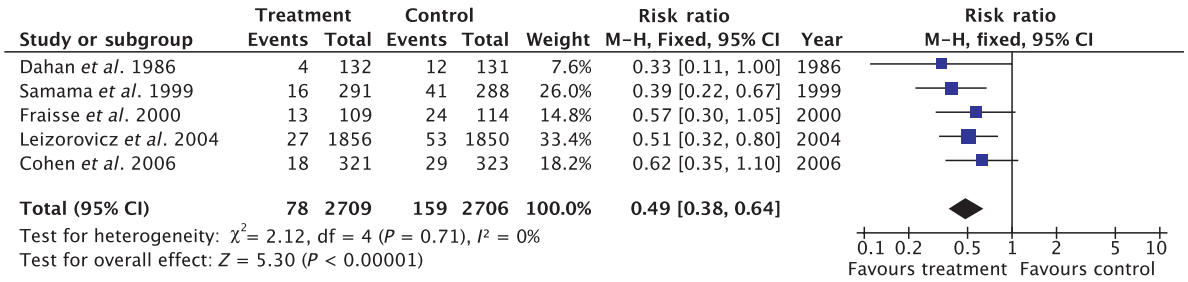

Fig. 1 Heparin prophylaxis versus control. Asymptomatic deep vein thrombosis during follow-up.

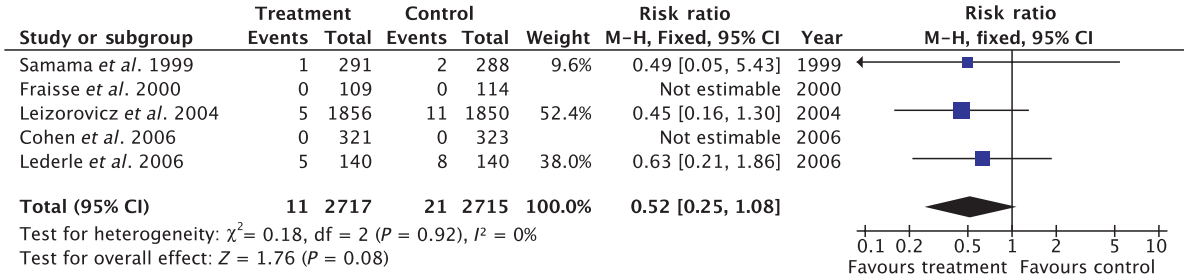

Fig. 2 Heparin prophylaxis versus control. Symptomatic deep vein thrombosis during follow-up.

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regarding abstracts from major meetings: International Society of Thrombosis and Haemostasis, American Society of Hematology, International Congress of Thrombosis and International Stroke Conference. Reference lists of identified publications were also scrutinized.

Study selection

Three independent investigators performed the study selection. Disagreements were resolved through discussion in the group. The criteria for study selection were:

- Published and abstracted randomized trials
- Acutely ill medical inpatients
- Comparison of fixed dose UFH, LMWH or fondaparinux with placebo
- Outcomes confirmed by objective diagnostic testing
- At least 1 week of prophylaxis and follow-up for VTE

Study data extraction

For each study, three independent investigators extracted data regarding study quality, population, intervention and outcome. We accepted the author's definitions of outcomes, and did not try to reclassify the events. Symptomatic DVT had to be confirmed by venography or compression ultrasound, pulmonary embolism (PE; Fig. 3) by ventilation-perfusion lung scanning, pulmonary angiography, helical computed tomography or autopsy. For major bleeding (Fig. 4) the study definition was accepted. Intracranial bleeding had to be confirmed by computed tomography, magnetic resonance imaging or autopsy.

The primary efficacy outcome was symptomatic VTE during the time of follow-up defined in each study. Data regarding on-treatment and after-treatment periods defined as secondary efficacy outcome in the original trials were not extracted. Primary safety outcome was defined as major bleeding, intracranial bleeding or fatal bleeding (Fig. 5).

Study or subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk ratio M-H, Fixed, 95% CI	Year
Dahan *et al*. 1986	1	132	3	131	8.6%	0.33 [0.03, 3.14]	1986
Samama *et al*. 1999	0	291	3	288	10.1%	0.14 [0.01, 2.73]	1999
Fraisse *et al*. 2000	0	109	0	114		Not estimable	2000
Leizorovicz *et al*. 2004	5	1856	6	1850	17.2%	0.83 [0.25, 2.72]	2004
Mahe *et al*. 2005	10	1230	17	1244	48.4%	0.59 [0.27, 1.29]	2005
Lederle *et al*. 2006	1	140	3	140	8.6%	0.33 [0.04, 3.17]	2006
Cohen *et al*. 2006	0	321	2	323	7.1%	0.20 [0.01, 4.18]	2006
Total (95% CI)	**17**	**4079**	**34**	**4090**	**100.0%**	**0.52 [0.29, 0.91]**	

Test for heterogeneity: $\chi^2 = 2.15$, df = 5 ($P = 0.83$), $I^2 = 0\%$
Test for overall effect: $Z = 2.29$ ($P = 0.02$)

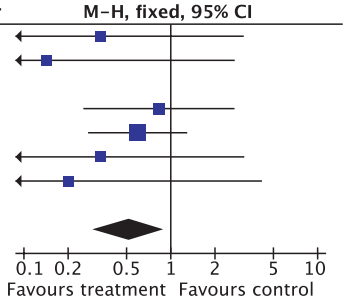

Fig. 3 Heparin prophylaxis versus control. Symptomatic pulmonary embolism during follow-up.

Study or subgroup	Treatment Events	Treatment Total	Control Events	Control Total	Weight	Risk ratio M-H, Fixed, 95% CI	Year
Dahan *et al*. 1986	1	132	2	131	10.9%	0.50 [0.05, 5.41]	1986
Samama *et al*. 1999	6	291	4	288	21.8%	1.48 [0.42, 5.21]	1999
Fraisse *et al*. 2000	6	109	3	114	15.9%	2.09 [0.54, 8.16]	2000
Leizorovicz *et al*. 2004	8	1856	0	1850	2.7%	16.95 [0.98, 293.36]	2004
Mahe *et al*. 2005	1	1230	3	1244	16.2%	0.34 [0.04, 3.24]	2005
Cohen *et al*. 2006	1	425	1	414	5.5%	0.97 [0.06, 15.52]	2006
Lederle *et al*. 2006	2	140	5	140	27.1%	0.40 [0.08, 2.03]	2006
Total (95% CI)	**25**	**4183**	**18**	**4181**	**100.0%**	**1.39 [0.77, 2.51]**	

Test for heterogeneity: $\chi^2 = 7.85$, df = 6 ($P = 0.25$), $I^2 = 23.6\%$
Test for overall effect: $Z = 1.08$ ($P = 0.28$)

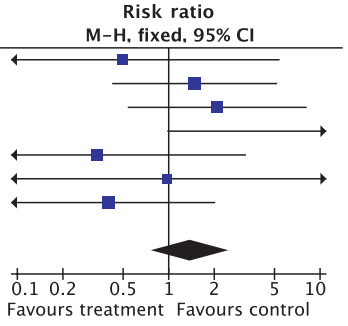

Fig. 4 Heparin prophylaxis versus control. Major bleeding during follow-up.

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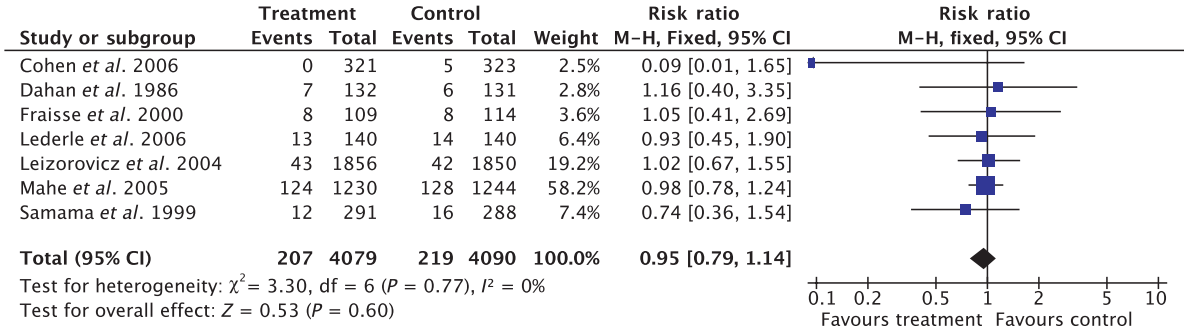

Fig. 5 Heparin prophylaxis versus control. All-cause death during follow-up.

Statistical analysis

Data were entered into the Cochrane Collaboration Review Manager package (RevMan 5). Heterogeneity between the studies was assessed with the chi-squared test and the chosen summary statistic variable was the reduction in relative risk (RR). The *P*-value for heterogeneity was above 0.10 in all analyses. The fixed model was used for calculations. A random-effects model was tested but showed similar results.

Results

We identified 10 studies assessing the prophylactic efficacy of heparin in acutely ill medical inpatients. Eight studies were blinded randomized controlled clinical trials, of which six used LMWH [6, 7, 10–13], one UFH [14] and one fondaparinux [8] as prophylaxis. The study using UFH [14] also included patients who had recently undergone surgery, and was therefore excluded. The remaining two studies [15, 16] were not blinded and compared UFH prophylaxis with no treatment. The study characteristics are presented in Table 1. The meta-analysis was performed only on the seven blinded randomized controlled clinical trials in nonsurgical patients using LMWH or fondaparinux to avoid selection bias or heterogeneity in the results due to different treatment regimens.

Outcomes

The main outcomes of LMWH prophylaxis or fondaparinux are summarized in Table 2. Asymptomatic DVT was by far the most common, with an incidence of 5.9% in the control group. This rendered an absolute risk reduction of 3.0%, number needed to treat (NNT) 33 to prevent one event. The incidence of symptomatic VTE was much lower, giving NNT of 271 for symptomatic DVT and 241 for symptomatic PE. Major bleeding according to the author's definitions was rare, slightly increased in the prophylaxis group giving a nonsignificant number needed to harm (NNH) of 598.

Discussion

A recent meta-analysis on anticoagulant prophylaxis for preventing VTE in hospitalized medical patients [9] used the same search criteria as in this study, but reached a different result. A meta-analysis is generally thought to be the golden rule, enlarging the existing data to reach as close to a reliable conclusion as possible. However, a meta-analysis is very sensitive to studies included, their heterogeneity and data extracted. In the field of anticoagulation prophylaxis to prevent VTE in acutely ill medical inpatients, most studies are small, use different heparins (UFH, various LMWHs, fondaparinux), different end-points (fibrinogen uptake test, ultrasound, phlebography, helical computed tomography, pulmonary scintigraphy, autopsy) and criteria vary for bleeding. The inclusion criteria vary extensively as well as treatment duration and time to follow-up. Thus, it is possible to extract different sets of data from the same group of studies depending on how end-points are selected for examination in the meta-analysis.

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Table 1 Study characteristics

Study, year	Patients (*n*)	Indication for prophylaxis	Prophylaxis regimen	Treatment duration	End-point, time	DVT diagnostics	PE diagnostics	Definition of major bleeding
Low-molecular weight heparin								
Dahan *et al.* [11]	270	Age >65, acute medical illness	Pharmuka 60 mg once daily	Until hospital discharge, ≤10 days	Asymptomatic DVT, <10 days	Fibrinogen uptake test	Not performed	Diffuse bleeding
Samama *et al.* [6]	1102	Age >40, Congestive heart failure (NYHA III-IV), acute or chronic respiratory disease, infectious or rheumatologic disease	Enoxaparin 20 mg or 40 mg once daily[a]	6–14 days	Asymptomatic DVT day 6 and day 14, symptomatic VTE ≤14 days	Venography screening	Scintigraphy Angiography Computed tomography	Hb drop ≥2 g dL^{-1}. Transfusion of 2 or more blood units. Retroperitoneal Intracranial. Fatal
Fraisse *et al.* [12]	223	Acute decompensated chronic obstructive pulmonary disease with mechanical ventilation	Nadroparin, 3800–5700E once daily	Until no longer mechanical ventilation, ≤21 days	VTE, ≤21 days	Ultrasound screening Venography	Angiography Computed tomography	Hb drop ≥2 g dL^{-1}. Transfusion of 2 or more blood units. Retroperitoneal Intracranial
Leizorovicz *et al.* [7]	3706	Age ≥40, projected hospitalization ≥4 days. Congestive heart failure (NYHA III-IV), acute or chronic respiratory disease, infectious or rheumatologic disease	Dalteparin 5000E once daily	14 days	VTE and sudden death, day 21	Ultrasound screening Venography	Scintigraphy Angiography Computed tomography Autopsy	Hb drop ≥2 g dL^{-1}. Transfusion of 2 or more blood units. Retroperitoneal Intracranial Spinal/epidural. Fatal. Need for substantial intervention
Mahe *et al.* [10]	2472	Age >40, hospitalized for <24 h, immobilized, acute medical illness	Nadroparin 7500E once daily	Until hospital discharge, ≤21 days	All-cause mortality, symptomatic VTE, ≤21 days	Venography Autopsy	Angiography Autopsy	Not described

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Table 1 (Continued)

Study, year	Patients (*n*)	Indication for prophylaxis	Prophylaxis regimen	Treatment duration	End-point, time	DVT diagnostics	PE diagnostics	Definition of major bleeding
Lederle *et al.* [13]	280	Age >60, hospitalization in general medical unit	Enoxaparin 40 mg once daily	Until hospital discharge	All-cause mortality, symptomatic VTE, day 90	Ultrasound Venography	Scintigraphy Angiography Autopsy	The bleeding severity index, Landefeld[24] 1989
Fondaparinux								
Cohen *et al.* [8]	849	Age ≥60, congestive heart failure (NYHA III-IV), acute respiratory, infectious or inflammatory disease, projected hospitalization ≥4 days	Fondaparinux 2.5 mg once daily	6–14 days	Asymptomatic VTE day 6 and day 15, symptomatic VTE ≤15 days	Venography screening	Scintigraphy Angiography Computed tomography Autopsy	Hb drop ≥2 g dL^{-1}. Transfusion of 2 or more blood units. Critical location. Need for surgical intervention. Fatal
Unfractionated heparin								
Belch *et al.* [15]	100	Heart failure, chest infection	Unfractionated heparin, 5000 U trice daily	Until mobilized	Asymptomatic VTE, ≤14 days	Fibrinogen uptake test	Not performed	Not described
Cade *et al.* [14]	234	Age >40, complete bed rest, cardiac failure, obesity, previous VTE, cancer or recent surgery	Unfractionated heparin, 5000 U twice daily	Until mobilized or ≤10 days	Asymptomatic VTE, ≤10 days	Fibrinogen uptake test	Not performed	Not described
Gårdlund [16]	11 693	Age >55, infectious disease Immobilization	Unfractionated heparin, 5000 U twice daily	Until hospital discharge, ≤21 days	Autopsy verified PE, ≤60 days	Autopsy	Autopsy	Serious bleeding diagnosed at autopsy

[a]Data from Enoxaparin 20 mg was not extracted [6].

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Table 2 All seven studies using low-molecular weight heparin (LMWH) or fondaparinux as prophylaxis [6–8, 10–13]. Events per included persons, frequency of symptomatic pulmonary embolism (PE), asymptomatic and symptomatic deep vein thrombosis (DVT), bleeding or death (%), absolute risk reduction (ARR) and numbers needed to treat/harm (NNT/NNH) are reported

	LMWH		Control			
	n	%	*n*	%	ARR (%)	NNT/NNH
Symptomatic PE	17/4079	0.42	34/4090	0.83	0.41	241
Symptomatic DVT	11/2717	0.40	21/2715	0.77	0.37	271
Asymptomatic DVT	78/2709	2.88	159/2706	5.88	3.00	33
Major bleeding	25/4183	0.60	18/4181	0.43	0.17	598

The same problem with validity of meta-analyses is also seen in the field of VTE prophylaxis in general surgery [17]. In this abstract five meta-analyses and two systematic reviews were compared. Despite substantial differences in design, these meta-analyses arrived at similar conclusions.

In the meta-analysis by Dentali *et al.* [9] the single largest study examining PE was only partly included [16], extracting data only from patients with PE during treatment with heparin. This study was excluded from our analysis as the comparison between UFH and no treatment was not blinded. This choice in study selection conveys the largest difference in results between the two meta-analyses. If this study is fully included, using the author's definition of time to follow-up, the effect of heparin on preventing PE is substantially reduced and no longer significant. This could suggest that the prophylactic effect of heparin in this setting is only effective during the treatment period, and that this effect will diminish over time. If so, the clinical benefit of short-term heparin prophylaxis would be small. An alternative explanation is that in this study UFH was used instead of the more recent drugs LMWH and fondaparinux. Randomized comparisons of enoxaparin with UFH have been performed in medical patients with heart failure or severe respiratory failure [18] and in acute ischaemic stroke [19] showing less efficacy of UFH.

There are differences between the two meta-analyses in terms of data extraction. The main difference is that Dentali *et al.* [9] chose to redefine the time of follow-up to be events during treatment with anticoagulation, while we used the time of follow-up predefined as primary outcome. We also excluded three events of sudden death that were regarded as PE in one study [8], as these events were not objectively confirmed by autopsy or radiology. Another small study with open-label use of UFH compared to no treatment was excluded by us but included by Dentali *et al.* [9]. Here they accepted two cases of PE that occurred 3 weeks after follow-up, i.e. not during anticoagulation treatment as defined for data extraction in their meta-analysis. Despite all these differences in study selection and data extraction, both meta-analyses show similar results. The main difference is in symptomatic PE, where Dentali *et al.* [9] show a highly significant preventive effect of 62%, compared to 48% in our study, that is also significant.

In our meta-analysis, LMWH prophylaxis in acutely ill medical patients prevented 48% of symptomatic PE, 48% of symptomatic DVT and 51% of asymptomatic DVT. As these end-points all are different manifestations of the same illness, VTE, one can argue that LMWH prophylaxis prevents approximately half of the expected VTE. There is a trend towards higher bleeding risk during LMWH prophylaxis (RR = 1.39), although not significant. Death is also not significantly affected by this treatment. NNT varies from 271 in symptomatic DVT to 33 in asymptomatic DVT, mainly due to a much higher incidence of the latter. In spite of extremely careful selection, the results in the individual studies had to be driven by asymptomatic VTE. Only in a meta-analysis is the small benefit on symptomatic VTE discerned. The absolute majority of the asymptomatic VTE were

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distal, and of questionable clinical significance. For the clinician, the important task is to identify those patients who have a sufficiently high risk for symptomatic VTE to warrant LMWH prophylaxis without an unnecessary risk of bleeding, i.e. a high net clinical benefit. Attempts have been made by risk scores [4] or registry searches [20] to identify groups of acutely ill medical inpatients at higher risk of developing VTE. Kucher *et al.* [21] identified a small group of medical inpatients with a high incidence of VTE, approximately 5% of all patients in the clinic. Among these, 80% had cancer which has to be considered as a strong risk factor for VTE. Other than cancer, VTE prophylaxis might be considered in immobile patients and in those with respiratory disease, previous VTE, sepsis, congestive heart failure or acute inflammatory disease [4, 20–23]. More studies are needed to create a better risk assessment for VTE that can guide the clinician in everyday practice. This is important, as PE commonly causes death in non-surgical patients too. However, in a meta-analysis, only an effect on morbidity, not mortality, is shown.

Conflict of interest statement

No conflict of interest was declared.

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